Subhankar Nag

Ph.D. in Computer Science and Engineering
Indian Institute of Technology Bombay, Mumbai, India
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RESEARCH INTERESTS

Generative models, Medical image computing, Neural image compression, Privacy-preserving machine learning

EDUCATION

Indian Institute of Technology Bombay, Mumbai, India — 2024 — 2029
Doctor of Philosophy in Computer Science and Engineering — CGPA: 9.13/10.00
Research focus: Deep Learning in Medical Image Computing

Ramakrishna Mission Vivekananda Educational and Research Institute, Howrah, India — 2021 — 2023
Master of Science in Computer Science — CGPA: 9.69/10.00
Thesis Title: Automatic Recognition and Generation of Indian Sign Language

Bankura Sammilani College, Bankura, India — 2018 — 2021
Bachelor of Science in Computer Science — CGPA: 9.19/10.00

PUBLICATIONS

Conference papers

- Subhankar Nag, Suyash P. Awate
  "How to train your generative diffusion model, when all training images are degraded, to model a distribution on high-quality images"
  IEEE International Conference on Image Processing (ICIP) 2026 (Accepted)

RESEARCH EXPERIENCE

Indian Institute of Technology Bombay (2024–Present)
Supervisor: Prof. Suyash P. Awate

- Investigating diffusion model training with only low-quality observations, for generating high-quality medical images
- Developing neural image representation for efficient compression of advanced MRI data
- Studying gradient inversion attacks and defense mechanisms to preserve the privacy in medical data

PROJECTS

Handwritten Video Frame Interpolation using Region-Specific Loss — 2024

- Implemented a CNN-based **RIFE** model with region-specific loss and extended input context
- Improved reconstruction quality from 37.7 dB to **40.9 dB PSNR (+3.2 dB)**

Brain Tumor Segmentation using Edge-Preserving Loss — 2024

- Implemented **UNet++** based model for brain tumor segmentation on the BraTS dataset
- Achieved **91.5 Dice score**, **improving performance by +1.6** using edge-based loss

Automatic Recognition and Generation of Indian Sign Language — 2023

- Developed **Transformer-based** and **3D-CNN-based** models for Indian Sign Language recognition
- Achieved **96.2% accuracy** on the INCLUDE dataset, **improving over baseline by +2.7%**

TEACHING EXPERIENCE

Indian Institute of Technology Bombay (2024-Present)

- Conducted post-class **doubt-solving sessions** for **Data Structures**, **Algorithms**, **C++**, and **AI/ML** courses, helping students understand the key concepts, clarifying problem-solving approaches, and assisting with assignments.

Achievements

GATE CS 2024: All India Rank **236** (out of 123,967 candidates)

SKILLS

- **Programming:** Python, C/C++, SQL
- **ML Frameworks:** PyTorch
- **Domains:** Medical Image computing, Diffusion Models, Deep learning